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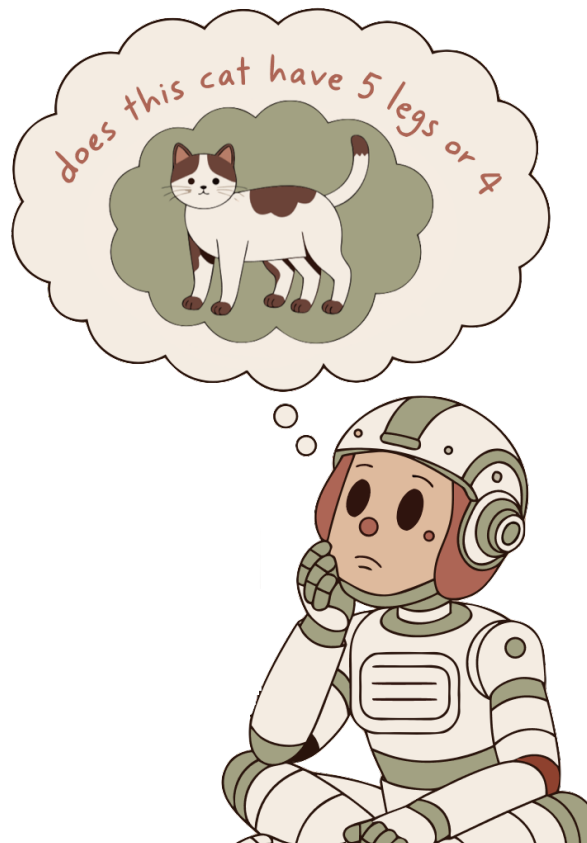
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VLMs Are Confidently Biased Before Producing the Final Answer

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Abstract

Vision-language models (VLMs) often fail on counterfactual images where visible evidence conflicts with familiar visual expectations. Existing benchmarks measure these failures through final-answer accuracy but provide limited insight into the probability distributions that produce them. We investigate counterfactual bias as a probability-level phenomenon using the VLMBias benchmark and three open-source VLMs, namely Qwen2.5-VL-3B, Qwen2.5-VL-7B, and LLaVA-OneVision-7B. Across models, the biased answer receives higher sequence probability in approximately 80% of counterfactual samples. Token-level analysis further shows that low-entropy answer decisions frequently favor the biased answer, indicating that models often commit confidently to prior-consistent responses before producing the final output. The analysis further reveals recurring patterns of prior substitution and full hallucination. To investigate whether these preferences remain dependent on visual evidence, we evaluate Visual Contrastive Decoding (VCD) as an inference-time intervention and diagnostic probe. Across most settings, VCD improves probability margins in fewer than half of samples, and its core assumption fails in the majority of cases. The main exception is the Flags domain, where VCD consistently improves margins and assumption-hold rates. Our findings suggest that counterfactual failures are characterized by strong prior-consistent probability preferences before the final answer is generated. They further indicate that VCD is most effective when bias remains visually gated but provides limited benefit when prior preferences already dominate the model’s predictions.

Code — <https://github.com/Rabu-Sufian/BiasInVLMs>

Datasets —

<https://huggingface.co/datasets/anvo25/vlms-are-biased>

1 Introduction

Large vision-language models (VLMs) combine visual perception with natural language generation, allowing them to answer questions, describe scenes, and reason about images through text (Liu et al. 2023; Bai et al. 2023). Although these models often perform well on standard visual inputs, their answers are not always fully grounded in the image itself. Many VLMs are built on pre-trained large language models

(LLMs), whose knowledge is learned from large-scale internet data. This gives VLMs broad prior knowledge about familiar objects, symbols, and visual patterns, but it can also make their answers depend on learned expectations rather than the specific evidence in the image (Sharma et al. 2024; Vo et al. 2025). Understanding when such prior-driven preferences dominate visual evidence, and whether they can be altered at inference time, remains an open challenge.

Such prior knowledge is often beneficial, helping models generalize and recognize familiar subjects. Problems arise when learned expectations conflict with visual evidence. Prior work suggests that LLMs can encode visual knowledge from text alone, including common object properties, spatial relations, and canonical facts such as the typical number of stripes on a flag or legs on an animal (Sharma et al. 2024). When inherited by VLMs, these expectations may influence predictions even when the image contradicts them.

Counterfactual images make this problem directly observable, as familiar subjects are modified in visually subtle but semantically important ways. A model may correctly recognize an ordinary Adidas shoe, national flag, or animal, yet fail when the same subject is altered in a way that contradicts common knowledge. The VLMBias benchmark directly tests this failure mode by modifying familiar visual subjects and asking neutral counting and identification questions (Vo et al. 2025). Unlike benchmarks that rely heavily on adversarial wording or yes/no prompts containing biased statements, VLMBias shifts the conflict into the image itself. The prompt remains relatively neutral, while the visual evidence contradicts the model’s prior expectations (Guan et al. 2024; Lee et al. 2025; Luo et al. 2025; Vo et al. 2025). This makes VLMBias particularly useful for studying whether VLMs answer from visual evidence or from memorized visual priors.

VLMBias reveals a large gap between ordinary recognition and counterfactual visual grounding. While the evaluated models perform reliably on unmodified images, their accuracy drops to 17.05% on counterfactual counting questions, with 75.70% of errors matching the predefined bias-aligned response (Vo et al. 2025). These results suggest that models often favor what they expect the image to show rather than what is visually present. However, VLMBias primarily evaluates this behavior through final-answer accuracy, providing limited insight into how strongly models

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prefer prior-consistent answers or whether those preferences remain amenable to inference-time intervention.

Building on VLMBias, this paper studies counterfactual VLM bias as a probability-level generation problem. Rather than treating the final answer as the primary unit of analysis, we examine the probability distributions that give rise to it during decoding. The analysis therefore uses margin-based diagnostics to compare the probability of the correct answer with the probability of the prior-consistent alternative and entropy-based diagnostics (Kuhn, Gal, and Farquhar 2023) to examine how confidently the model commits to its answer during generation. Together, these diagnostics make it possible to study not only whether a VLM fails, but how prior-driven preferences appear in probability distributions and how strongly the model favors the prior-consistent response.

The paper further investigates whether inference-time contrastive decoding can alter these probability-level preference dynamics. Visual Contrastive Decoding (VCD) is a training-free method that contrasts output distributions derived from original and visually distorted images in order to suppress language priors and amplify visually grounded information (Leng et al. 2024). Originally proposed to mitigate object hallucination, VCD operates under the assumption that visual degradation increases reliance on language priors, making the contrast between original and distorted-image distributions informative (Leng et al. 2024). In this work, VCD is used both as an intervention and as a diagnostic probe. Under visually gated bias, VCD should shift probability toward the correct answer. If the underlying preference is largely insensitive to the visual signal, however, the contrast may be uninformative and the intervention may fail.

This paper is guided by two research questions:

1. **RQ1:** How do prior-driven probability preferences manifest during counterfactual visual reasoning, and how do these dynamics vary across models and domains?
2. **RQ2:** Does inference-time contrastive decoding alter these probability-level preference dynamics, and under what conditions does it fail to do so?

Our analysis reveals that all three models assign higher sequence log-probability to the biased answer in approximately 80% of counterfactual samples, with mean margins of -1.31 , -3.65 , and -2.78 for Qwen2.5-VL-3B, Qwen2.5-VL-7B, and LLaVA-OV-7B, respectively. Token-step diagnostics reveal two recurring failure patterns. In prior substitution, the model commits confidently to a memorized canonical value despite contradictory visual evidence. In full hallucination, uncertainty resolves toward a third response dominated by neither candidate. Applied as a diagnostic probe, VCD helps fewer than 50% of samples in five of six model-distortion conditions. Its central assumption, that distortion increases reliance on language priors, holds in fewer than 40% of cases for most configurations. The exception is Flags, where assumption-hold rates and intervention success both exceed 61%, motivating a distinction between preferences that remain responsive to visual evidence and those that do not.

2 Related Work

The failure mode studied in this paper is connected to hallucination in vision-language models (VLMs), but it is narrower than standard object hallucination. In object hallucination, a model produces a plausible answer that conflicts with the visual evidence, for example by mentioning or confirming an object that is not present. POPE (Li et al. 2023) formalizes this problem as an object-presence task, making hallucination measurable but largely restricting evaluation to yes/no recognition settings. MME (Fu et al. 2025) broadens the scope to perception and cognition subtasks, including existence, counting, position, color, OCR, and commonsense reasoning. HallusionBench (Guan et al. 2024) further suggests that hallucination can emerge from interactions between visual evidence and language-based expectations rather than from object recognition failure alone. Across these benchmarks, evaluation is conducted at the level of final outputs; the question of how prior knowledge and visual evidence compete during generation remains largely unaddressed.

At the same time, not every visual error should be attributed to prior bias. Strong VLMs can also fail on low-level visual tasks even when no familiar object prior is involved. *Vision Language Models Are Blind* (Rahmanzadehgervi et al. 2025) reports failures on overlap detection, line intersection, and counting visual elements, while ORBIT (Kolari et al. 2026) finds weaknesses in object-property reasoning under realistic and counterfactual conditions. These findings establish perceptual reliability as a necessary baseline condition before interpreting failures as prior substitution. However, other work shows that language models can encode substantial visual knowledge from text alone, including object properties, spatial relations, and canonical visual facts (Sharma et al. 2024). VLind-Bench (Lee et al. 2025) evaluates language priors while controlling for knowledge and perception confounds. ViLP (Luo et al. 2025) studies prior-consistent behavior using out-of-distribution images and structured answer-image triples. *Words or Vision* (Deng et al. 2025) further examines cross-modal conflict and finds that VLMs often over-trust textual information when language and image disagree. These works collectively motivate evaluation settings where visual evidence is intentionally placed in conflict with familiar learned expectations.

Counterfactual images provide one way to test whether a VLM answers from visual evidence or learned associations. VLMBias (Vo et al. 2025) is the benchmark most closely related to the setting studied here. Its core contribution is to introduce counterfactual modifications directly into familiar visual structures in order to evaluate whether model responses remain grounded in visible evidence or instead revert to dominant learned priors. To account for variability across visual concepts and reasoning settings, the benchmark spans multiple domains including animals, logos, flags, chess boards, game boards, optical illusions, and patterned grids. Additional analyses in VLMBias show that background cues can strengthen prior-consistent answers, that helpful prompts provide only limited improvements, and that subject labels embedded within the image can further increase biased responses (Vo et al. 2025).

Probability diagnostics

Most VLM hallucination and bias benchmarks evaluate final outputs using metrics such as accuracy, hallucination rate, or bias rate. While these measures are useful for comparing models, they provide limited insight into how an answer forms during autoregressive generation. In the broader language-model literature, probability-based signals are commonly used to study confidence, uncertainty, and generation dynamics. Calibration work examines whether predicted probabilities correspond to empirical correctness (Guo et al. 2017), while Kadavath et al. (Kadavath et al. 2022) use model probabilities and self-evaluation signals to estimate whether a model knows an answer. Semantic entropy further extends this perspective by measuring uncertainty over meanings rather than surface forms (Kuhn, Gal, and Farquhar 2023).

These approaches motivate probability-level analysis as a way to study how models commit to answers during generation. In particular, changes in log-probability margins and entropy can reveal patterns such as early commitment, confidence concentration, or entropy collapse before a final answer is produced. More recent inference-time methods similarly operate directly on decoding distributions in order to reduce hallucination and strengthen visual grounding (Leng et al. 2024; Huang et al. 2024; Huo et al. 2025; Wang et al. 2024). However, these approaches have primarily been studied in general hallucination settings rather than in counterfactual visual-prior conflicts.

The same output distributions are also the target of inference-time mitigation methods. Visual Contrastive Decoding (VCD) (Leng et al. 2024) modifies decoding distributions using contrasts between original and visually distorted inputs in order to suppress generations driven by statistical bias or unimodal language priors. Related methods also intervene during decoding. OPERA reduces multimodal hallucination by penalizing over-trust in summary tokens (Huang et al. 2024), while Self-Introspective Decoding and instruction-based contrastive decoding modify decoding distributions to reduce hallucination without additional training (Huo et al. 2025; Wang et al. 2024). These methods show that decoding can influence hallucination and grounding, but they have mostly been evaluated on hallucination benchmarks rather than counterfactual visual-prior conflicts.

3 Dataset

This study uses the publicly released VLM-Bias dataset, available on Hugging Face as `anvo25/vlms-are-biased` (Vo, Nguyen, and collaborators 2025). The dataset was produced as part of the VLMBias benchmark pipeline, which constructs controlled counterfactual visual examples by modifying familiar subjects while keeping them recognizable (Vo et al. 2025). All generated images underwent manual human review, with low-quality or ambiguous samples rejected prior to release. We use the released dataset directly, without regenerating images or altering labels.

Two splits are used in this paper: the *Original Split* and the *Main Split*. The Original Split contains unmodified images

paired with counting and identification questions. It serves as a control condition for visual competence because the correct answer is expected to agree with the canonical version of the subject. The Main Split contains minimally modified counterfactual images and serves as the primary setting for prior-conflict analysis. In this split, the model must recognize a familiar subject while updating its answer according to the modified visual evidence.

The Optical Illusions domain is excluded from both splits because it is intentionally designed to create perceptual ambiguity, including for human observers. Since this paper studies prior-driven model failures rather than ambiguity in human perception, including this domain would introduce an additional confound. Patterned Grid appears only in the Main Split because it is a constructed anomaly task rather than a modified version of a familiar original subject.

Rows correspond to image-prompt pairs rather than unique images. In the Main Split, each base image is paired with two neutral counting prompts and provided at three resolutions: 384, 768, and 1152 pixels. We therefore distinguish between row counts and unique base-image counts in Fig. 1. After excluding Optical Illusions, the dataset contains 326 rows and 163 base images in the Original Split, and 1,992 rows and 332 base images in the Main Split. The token-level subset contains 154 rows for the Original Split and 1,674 rows for the Main Split.

Answer Fields

Each Main Split sample contains two answer fields used in the analysis: `ground_truth` and `expected_bias`. The `ground_truth` field gives the visually correct answer for the counterfactual image, while `expected_bias` gives the prior-consistent answer associated with the canonical version of the subject. This structure makes it possible to distinguish visually grounded answers from prior-aligned errors and other mistakes.

The Original Split does not contain a true bias-aligned answer because the unmodified image is expected to agree with the canonical prior. For probability comparison in this control condition, we therefore define a *proxy alternative*. This proxy is not treated as a bias label; it is used only to compare the model’s preference for the correct answer against a nearby numeric alternative. The construction of the proxy alternative is described in Section 4.

4 Methodology

This paper studies counterfactual VLM bias through probability-level preference dynamics during answer generation. The methodology is organized into two phases. Phase 1 analyzes prior-driven preference using two complementary diagnostics. Sequence-level scoring evaluates predefined candidate answers through direct probability scoring, while token-step analysis inspects the vocabulary distribution at the first generated answer token during free-form decoding. Phase 2 evaluates whether Visual Contrastive Decoding (VCD) (Leng et al. 2024) can shift these probability-level preferences toward visually grounded answers. Candidate probabilities are compared across original and distorted

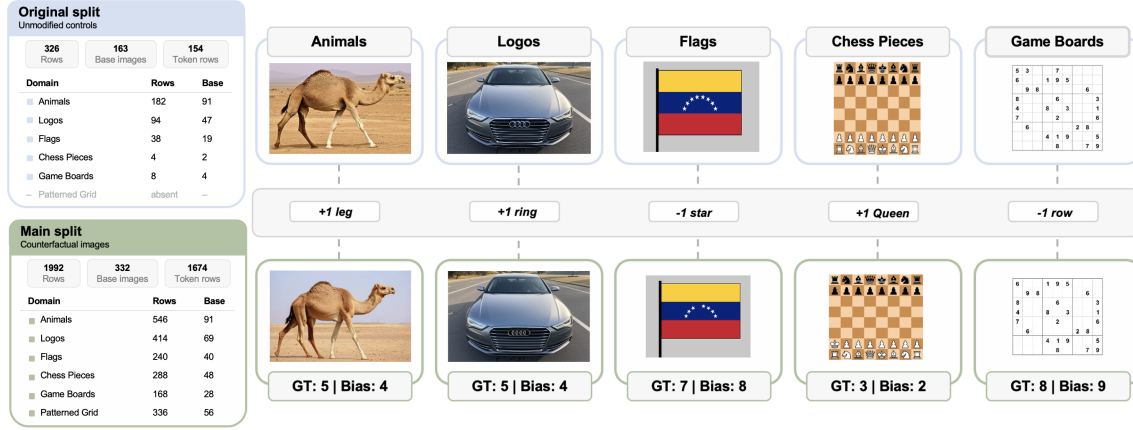


Figure 1: Overview of the VLMBias dataset used in this study after excluding Optical Illusions. Rows correspond to image-prompt pairs, while base images collapse repeated prompts and, for the Main Split, resolution variants. Token rows indicate the subset used for token-level probability analysis.

image conditions to quantify intervention effects and characterize failure cases.

Model Selection

To support this design, we evaluate three open-source vision-language models: Qwen2.5-VL-3B-Instruct, Qwen2.5-VL-7B-Instruct (Wang et al. 2025), and LLaVA-OneVision-7B (Li et al. 2024). The two Qwen2.5-VL models share the same model family but differ in scale, allowing a direct comparison between 3B and 7B parameter variants. LLaVA-OneVision-7B is included as an architecture-level comparison at a similar scale to Qwen2.5-VL-7B. This selection allows the analysis to examine whether counterfactual bias patterns are mainly affected by model scale, architecture, or shared multimodal training behavior.

All selected models provide access to token-level logits, which is necessary for the probability-based diagnostics used in this study. In addition, all three models use the Qwen2 tokenizer family, reducing vocabulary-related variation during token- and sequence-level probability comparisons.

Prompting and Inference Setup

All experiments use the prompts provided by the VLMBias dataset in order to preserve comparability with the benchmark setup. VLMBias uses neutral visual questions (Vo et al. 2025) that fall into two main categories: counting prompts and identification prompts.

Inference is performed using deterministic decoding wherever free-form generation is required. Sampling is disabled to avoid variation caused by stochastic decoding. For each run, the generated answer, extracted answer, token sequence, and model logits are stored. The extracted answer is compared against the relevant answer fields described in Section 3.

Token-step analysis requires the model to produce the answer immediately so that the first answer-token distribution

can be inspected. For this part of the analysis, the following instruction is appended to the original prompt:

Answer with exactly ONE digit (0–9). No text, no multiple digits.

This constrained format is used only for token-step diagnostics. Sequence-level scoring does not require it, as candidate answers are scored directly by computing their log probability conditioned on the full input. For token-step analysis, `max_new_tokens` is set to 3, which is sufficient for any single-digit response plus minimal surrounding tokens. All models are loaded with `float16` precision and automatic device mapping, with settings kept fixed across models, splits, and experimental conditions.

Phase 1: Probability-Level Analysis

Phase 1 analyzes prior-driven preference through log-probability margins and entropy, measuring preference direction and prediction confidence at the answer step across models and domains. The full Phase 1 pipeline, including the sequence-level and token-step branches, is illustrated in Appendix A.

Sequence-Level Candidate Scoring Sequence-level scoring evaluates predefined candidate answers by computing their log probability under the model, conditioned on the image and prompt. For each sample, the visually correct candidate is denoted as *GT*. In the Main Split, the prior-consistent candidate is denoted as *bias*. In the Original Split, where no true bias-aligned answer exists, this candidate is replaced by the proxy alternative described in Section 4.

For a candidate answer $a = (a_1, \dots, a_T)$, the full input is constructed by applying the model-specific chat template to the image and prompt and then appending the candidate answer. The sequence is passed through the model in a single forward pass under teacher forcing (Sutskever, Vinyals, and Le 2014), where the candidate answer tokens are provided directly during scoring rather than generated autore-

gressively. The prompt is tokenized separately to identify the answer-token span, and log probabilities for answer tokens are extracted from the model logits using log-softmax.

Two score representations are computed for each candidate:

$$\ell_{\text{seq}}(a) = \sum_{t=1}^T \log p(a_t \mid x, v, a_{<t}) \quad (1)$$

$$\ell_{\text{avg}}(a) = \frac{1}{T} \ell_{\text{seq}}(a) \quad (2)$$

where x is the prompt, v is the image, and T is the number of answer tokens. Since candidate answers can differ in token length, ℓ_{avg} is used as the primary score representation. For the 96.7% of samples where candidate token lengths are identical, ℓ_{seq} and ℓ_{avg} are numerically equivalent.

Raw candidate probabilities may partly reflect lexical or numeric frequency effects independent of image content. To reduce this confound, each candidate is additionally scored using the context-free prompt "Answer: ". Following calibration-based correction of language priors (Zhao et al. 2021), the PMI-corrected score is:

$$\text{PMI}(a) = \ell_{\text{avg}}(a \mid x, v) - \ell_{\text{avg}}(a \mid \text{"Answer: "}) \quad (3)$$

The PMI-adjusted margin $M_{\text{PMI}} = \text{PMI}(\text{gt}) - \text{PMI}(\text{bias})$ serves as a robustness check: agreement in sign between M_{PMI} and M_{seq} indicates that the observed preference is not explained by lexical frequency or tokenization artifacts.

Proxy Alternative Construction As described in Section 3, the Original Split does not contain a predefined bias candidate. To make probability comparisons possible in this control condition, we construct a proxy alternative for numeric counting rows.

For each Original Split sample, all integer candidates in the window $[\text{GT} - 2, \text{GT} + 2]$ are scored using the same teacher-forced sequence scoring procedure described in Section 4. The proxy alternative is selected as the highest-scoring non-ground-truth candidate according to average log probability:

$$\hat{a}_{\text{proxy}} = \arg \max_{a \neq \text{GT}, a \in [\text{GT}-2, \text{GT}+2]} \ell_{\text{avg}}(a \mid x, v). \quad (4)$$

This candidate is not interpreted as a bias label. It serves only as a nearby numeric reference point, allowing the same margin and entropy diagnostics used on the Main Split to be applied to the Original Split. The construction is illustrated in Fig. 2.

Margin and Entropy Metrics The primary outcome variable is the sequence-level margin:

$$M_{\text{seq}} = \ell_{\text{avg}}(\text{GT}) - \ell_{\text{avg}}(\text{bias}) \quad (5)$$

A positive margin indicates preference toward the visually correct answer; a negative margin indicates preference toward the prior-consistent or proxy alternative. Aggregate performance is reported as *% prefers gt*, defined as the proportion of samples where $M_{\text{seq}} > 0$.

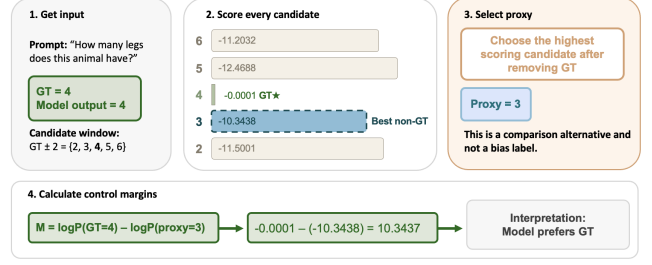


Figure 2: Proxy alternative construction for an example Original Split sample. All candidates in the $\text{GT} \pm 2$ window are scored independently; the highest-scoring non-ground-truth candidate is selected as the proxy alternative.

Margins indicate which candidate the model prefers, but not how concentrated the model’s distribution is. Entropy is therefore computed over the vocabulary distribution at each answer-token position:

$$H_t = - \sum_{w \in V} p(w \mid x, v, a_{<t}) \log p(w \mid x, v, a_{<t}) \quad (6)$$

where V is the model vocabulary. Two summary measures are used. First-step entropy, H_{step1} , is computed at the first answer-token position, while mean entropy is computed across all generated answer tokens as:

$$H_{\text{mean}} = \frac{1}{T} \sum_{t=1}^T H_t \quad (7)$$

The point-biserial correlation between *% prefers gt* and H_{step1} is computed per model to examine whether biased preference is associated with lower prediction uncertainty at the answer step. A negative margin with low entropy suggests confident prior substitution; a negative margin with high entropy suggests a weaker or more uncertain preference.

Token-Step Analysis Token-step analysis complements sequence-level scoring by examining the probability distribution during free-form generation. It focuses on samples where both the ground-truth and bias or proxy answers are represented as single tokens under the model tokenizer. This restriction avoids silently truncating multi-token candidates and ensures that both candidates are compared within the same vocabulary distribution.

For each valid sample, generation is run with `output_scores=True` and `return_dict_in_generate=True`, retaining the full vocabulary distribution at each generation step. The generated token sequence is scanned from left to right to identify the first step s at which a digit token appears. The vocabulary distribution at this step is used to extract the log probabilities of the GT and bias tokens. The token-step margin is defined as:

$$M_{\text{tok}} = \log p(t_{\text{GT}} \mid x, v, y_{<s}) - \log p(t_{\text{bias}} \mid x, v, y_{<s}). \quad (8)$$

A positive token-step margin indicates preference for the visually correct token at the answer step, while a negative margin indicates preference for the bias or proxy token. Samples with multi-token candidates, missing answer steps, or non-numeric generated outputs are excluded from token-step analysis. These exclusions affect only the token-level subset and do not remove the same samples from sequence-level scoring. For the Original Split, the proxy bias candidate described in Section 3 is substituted for `expected_bias` when computing `margin_step`.

Phase 2: VCD as an Intervention

Phase 2 applies Visual Contrastive Decoding (VCD) (Leng et al. 2024) over distorted image conditions using the same probability-based candidate scoring framework established in Phase 1. Analysis focuses on how distortion alters candidate log probabilities and whether contrastive decoding changes preference direction across models, domains, and distortion types. A full overview of the VCD pipeline is provided in Appendix B.

Distortion Setup VCD requires a distorted version of each image to construct the contrastive distribution used during decoding. Two distortion types were applied independently across all models and samples.

Gaussian noise was applied by sampling pixel-wise perturbations from $\mathcal{N}(0, \sigma^2)$ with $\sigma = 100$, added to each RGB channel and clipped to $[0, 255]$. A fixed random seed was used to ensure reproducibility across models. Gaussian blur was applied using a filter with radius $r = 10$, removing high-frequency spatial detail while preserving coarse image structure visible to a human observer. Both parameter values were selected to ensure sufficient distortion for the contrastive signal without fully obscuring the image content.

For each sample, both GT and bias candidates were scored under the original image and the distorted image condition. This produced paired real and noisy log-probability estimates for each candidate across both distortion types.

VCD-Adjusted Margins Given the real and distorted log-probability scores for each candidate, the VCD-adjusted score for a candidate a is defined as:

$$\ell_{\text{vcd}}(a) = (1 + \alpha) \ell_{\text{avg}}(a \mid \text{real}) - \alpha \ell_{\text{avg}}(a \mid \text{distorted}) \quad (9)$$

where α controls the strength of the contrastive correction. The VCD-adjusted margin is then defined as:

$$\text{vcd.margin} = \ell_{\text{vcd}}(\text{GT}) - \ell_{\text{vcd}}(\text{bias}) \quad (10)$$

The formulation increases the contribution of the real-image score while subtracting the corresponding distorted-image score for each candidate. Three values of $\alpha \in \{0.5, 1.0, 2.0\}$ are evaluated to assess sensitivity to correction strength, with $\alpha = 1.0$ used as the primary reported setting. The unadjusted `real_avg_margin` from Phase 1 serves as the baseline for comparison.

To quantify the effect of distortion independently of contrastive decoding, two auxiliary shift metrics are additionally computed per candidate:

$$\text{gt_lp_shift} = \ell_{\text{avg}}(\text{gt} \mid \text{distorted}) - \ell_{\text{avg}}(\text{gt} \mid \text{real}) \quad (11)$$

$$\text{bias_lp_shift} = \ell_{\text{avg}}(\text{bias} \mid \text{distorted}) - \ell_{\text{avg}}(\text{bias} \mid \text{real}) \quad (12)$$

Positive values indicate that a candidate receives higher probability under the distorted image condition than under the original image condition.

Interpreting Intervention Success and Failure VCD intervention success is evaluated at the sample level. A sample is classified as *helped* when the VCD-adjusted margin exceeds the unadjusted real-image margin:

$$\text{helped} = (\text{vcd.margin}_{\alpha=1} > \text{real_avg_margin}) \quad (13)$$

The proportion of samples satisfying this condition is reported as *% helped*, aggregated per model, distortion type, and topic.

To separate different intervention outcomes, two auxiliary conditions are tracked using the distortion-induced log-probability shifts defined in the previous subsection. The first condition, termed *assumption holds*, is defined as:

$$\text{bias_lp_shift} > 0 \quad (14)$$

indicating that the bias candidate receives higher probability under the distorted image condition than under the original image condition.

Two intervention failure categories are then distinguished. The first, *assumption failure*, occurs when $\text{bias_lp_shift} \leq 0$. The second, *correction failure*, occurs when $\text{bias_lp_shift} > 0$ but *helped* remains false. The rate of *assumption holds* is reported alongside *% helped* to separate these two intervention outcomes.

5 Analysis

Before we interpret Main Split errors as counterfactual bias, we first test whether the models can solve the same tasks on unedited images. The Original Split serves as this control condition. In this split, the visible image and the expected prior agree. A visually grounded model should therefore usually prefer the ground truth answer.

All three models perform substantially above chance on the Original Split. Under raw exact matching, overall accuracy is 69.3% for LLaVA-OV-7B, 77.9% for Qwen2.5-VL-3B, and 78.5% for Qwen2.5-VL-7B. Because identification questions are sensitive to harmless lexical variation, we also report corrected scores. Under the broader visual correction, overall accuracy increases to 77.3% for LLaVA-OV-7B, 84.7% for Qwen2.5-VL-3B, and 86.5% for Qwen2.5-VL-7B (See Appendix C.1). These results indicate that the models are not generally incapable of the underlying visual tasks.

The topic-level breakdown shows that Original Split competence is domain-dependent, but these subgroup results should be interpreted cautiously because the split is small and unevenly distributed across domains. After excluding

Optical Illusions, the Original Split contains 326 image-prompt rows and 163 base images, with some domain-task combinations represented by only a small number of examples. As a result, subgroup percentages are used as qualitative diagnostics of where the control condition is cleaner or noisier, rather than as stable estimates of model performance.

With this caveat, Animal counting is the cleanest control condition, with all three models reaching 100% accuracy. This confirms that the models can solve ordinary animal-leg counting when the visual evidence agrees with the canonical prior. This point is important for the counterfactual analysis, since failures on counterfactual animal images cannot be explained by a general inability to count animal legs in the original setting.

Other domains are less stable. Flag counting is already weak on original images, with accuracies of 36.8% for LLaVA-OV-7B, 57.9% for Qwen2.5-VL-3B, and 42.1% for Qwen2.5-VL-7B (See Appendix C.2). Counterfactual flag results must therefore be interpreted relative to a noisier visual baseline. The models often recognize flags in identification form, but exact counting of visual elements such as stars remains difficult. Logos show a different pattern, with near-perfect identification performance but substantially lower counting accuracy. Qwen2.5-VL-7B performs best at 74.5%, followed by LLaVA-OV-7B at 66.0% and Qwen2.5-VL-3B at 57.4%. These errors indicate that some domains contain visual counting difficulty even without counterfactual edits, and these baselines are therefore used to contextualize the Main Split results rather than treating all counterfactual failures as equally diagnostic of prior bias. Full Original Split results are reported in Appendix C.

Prior-Driven Bias and Failure Modes

Sequence scoring shows a clear reversal between the Original Split and the Main Split. On original images, the models usually assign higher probability to the ground truth count than to the strongest nearby proxy answer. On counterfactual images, this pattern collapses. The ground truth answer receives higher sequence probability than the prior consistent answer in only 19.4% of samples for Qwen2.5-VL-3B, 15.3% for Qwen2.5-VL-7B, and 16.9% for LLaVA-OV-7B. In the Original Split control, the same models prefer the ground truth in 72.4%, 90.2%, and 84.7% of samples.

The margin values show that the models are not simply close to the decision boundary. Mean `avg_margin` values of -1.31 , -3.65 , and -2.78 indicate that models are not hovering near the decision boundary; rather, they are assigning substantially higher probability to the wrong answer. Qwen2.5-VL-7B’s margin is nearly three times larger in absolute value than Qwen2.5-VL-3B’s, suggesting that the larger model commits to its prior more forcefully, not less. LLaVA-OV-7B exhibits similarly strong negative margins despite architectural differences, indicating that the behavior is not isolated to a single model family.

Token-step scoring supports the same conclusion. On the single-digit subset, generated correct answer rates are only 9.0%, 10.6%, and 6.8% for Qwen2.5-VL-3B, Qwen2.5-VL-7B, and LLaVA-OV-7B. This shows that the effect is not

only produced by candidate scoring. The prior preference is already visible at the first answer token during generation. A small number of rows were excluded from token step analysis because the model produced a multi-digit answer even though the ground truth was a single digit. The exclusion rate is 8.1% for Qwen2.5-VL-3B, 0.8% for Qwen2.5-VL-7B, and 0.5% for LLaVA-OV-7B. These exclusions do not change the overall pattern.

Domain Level Breakdown The aggregate failure rate obscures meaningful variation across domains. Not all counterfactual manipulations are equally at odds with model priors. Figure 3 shows how often each model assigns higher probability to the ground truth answer than to the prior consistent answer, while we in Figure 4 break down the corresponding mean probability margins.

Animals is the clearest case of prior dominance. All three models almost always prefer the canonical answer over the visible counterfactual answer. Qwen2.5-VL-3B has a mean margin of -3.78 , Qwen2.5-VL-7B has a mean margin of -8.13 , and LLaVA-OV-7B has a mean margin of -6.57 . Even after PMI correction, the domain stays near zero ground truth preference for all three models, suggesting that these cases are unlikely to be explained by answer frequency alone. This result is especially informative, as animal counting is perfect in the Original Split. The models can count ordinary animal legs when the image agrees with the prior. They fail when the visible number of legs contradicts the usual animal count. This supports the interpretation that the Main Split error is driven by prior override rather than general visual counting failure.

Chess Pieces also shows strong counterfactual failure. Ground truth preference is almost zero for Qwen2.5-VL-3B and Qwen2.5-VL-7B, and only 2.1% for LLaVA-OV-7B. The mean margin is most negative for Qwen2.5-VL-7B at -4.01 . PMI correction leaves the two Qwen models unchanged, but increases LLaVA-OV-7B to 14.6% (see Figure 3). This suggests that LLaVA’s raw chess result is partly affected by candidate preference, but the domain remains far from visually grounded. This shows a strong prior pressure. At the same time, the generated outcome pattern is more model dependent than in Animals. Qwen2.5-VL-7B often produces the prior consistent answer, while Qwen2.5-VL-3B and LLaVA-OV-7B more often produce other wrong numeric answers. Chess Pieces combines prior pressure with instability in the final generated answer.

Flags is the main contrast case. In Figure 3, Flags is the only domain where ground truth preference approaches or exceeds 50% for two models, and the margins sit close to zero. PMI changes little, so the result is not mainly a frequency artifact. However, the Original Split showed that flag counting is already noisy (Appendix C.2). Flags reflects weaker prior dominance, not reliable visual grounding. Logos, Game Boards, and Patterned Grid are more mixed. Their margins are closer to zero, but the models still fail often. Larger PMI shifts in these domains suggest a mixture of weaker priors, candidate preference, and unstable numeric reasoning. Logos also shows a mixed pattern. Ground truth preference is low, but PMI adjustment changes the estimate

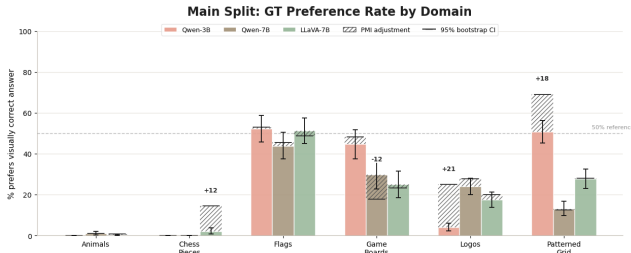


Figure 3: Share of counterfactual samples where each model assigns higher sequence probability to the ground truth answer than to the prior consistent answer. Error bars show 95% bootstrap confidence intervals. Hatched bars show the PMI adjusted estimate.

more than it does for Animals. This suggests that part of the raw logo result may reflect answer frequency or candidate preference rather than only image based prior bias. Even after this adjustment, Logos remains far from reliable visual grounding.

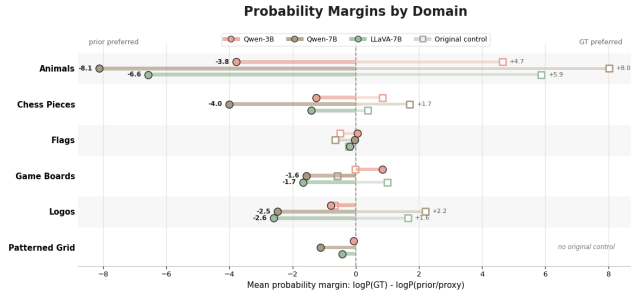


Figure 4: Mean probability margin by domain. Positive values mean the model prefers the ground truth answer. Negative values mean the model prefers the prior consistent answer. Square markers show the Original Split control where available. Patterned Grid has no Original Split control.

Margin and Entropy Diagnostics The aggregate bias rates reported above establish that the models fail, but not how they fail. Margin and entropy signals at the token level provide a more granular view, specifically whether the model is uncertain and erring, or confident and wrong.

The Original Split provides the expected control pattern (Appendix C.3). On unmodified images, low-entropy answer-step distributions strongly favour the ground-truth token. In other words, when the models are confident and no counterfactual conflict is present, confidence is usually visually grounded. The Main Split reverses this relationship. Under token-step scoring on counterfactual images, entropy at the decision step correlates strongly with margin across all three models (Qwen2.5-VL-3B: $r = 0.823$, Qwen2.5-VL-7B: $r = 0.734$, LLaVA-OV-7B: $r = 0.874$). As shown in Figure 5, samples where the model prefers the biased answer tend to exhibit low entropy. The model is not wavering between two plausible interpretations; it is often committing confidently to the wrong one.

This reversal is important. The models are not simply more uncertain on counterfactual images. Instead, the counterfactual manipulation changes what the model is confident about. On original images, confidence favors the ground-truth answer, whereas on counterfactual images it favors the prior-consistent answer. High-entropy Main Split samples are more likely to yield positive margins, suggesting that uncertainty at the decision step is one of the few conditions under which visual evidence can compete with the learned prior. This fits the domain results above. Strong prior domains such as Animals and Chess Pieces do not only produce low ground truth preference. They also produce confident prior preference.

This relationship does not survive under sequence scoring, where the same correlation effectively vanishes (Qwen2.5-VL-3B: $r = 0.057$, Qwen2.5-VL-7B: $r = 0.223$, LLaVA-OV-7B: $r = 0.219$). The divergence between the two methods is informative. Sequence scoring integrates log-probabilities across the full answer sequence, smoothing over the sharp commitment that occurs at a single decision step. The strong token-level correlation suggests that entropy collapse is a generation-time phenomenon. It emerges at the precise moment the model commits to an answer, rather than representing a stable property of the full candidate sequence.

Two Failure Modes Grouping domains by their joint margin and entropy signatures reveals two qualitatively distinct failure modes, each associated with a different relationship between prior preference and visual evidence. Figure 6 illustrates this separation across domains and models under token-step scoring.

The first mode, we define as *prior substitution*, characterises Animals and Chess Pieces. In this case, the model outputs the canonical answer instead of the visible ground truth. Animals is our clearest example here. Prior substitution accounts for 93% of Qwen2.5-VL-3B animal outputs and 99% of both Qwen2.5-VL-7B and LLaVA-OV-7B animal outputs. This matches the earlier margin result, where Animals had the strongest negative probability margins. The probability and generation results show a clean form of prior override. The model does not merely fail to count the image. It replaces the visible count with the expected canonical count.

The second mode is *full hallucination*. In this case, the model does not output the ground truth or the predefined prior answer. It generates a third number. This is most visible in Patterned Grid for Qwen2.5-VL-3B and Qwen2.5-VL-7B, where full hallucination accounts for 74% and 60% of outputs. It is also visible in Chess Pieces for Qwen2.5-VL-3B and LLaVA-OV-7B. This shows why probability margins alone are not enough. A model can show prior pressure in its scores but still produce a different wrong answer during generation.

Several domains mix the two modes. Logos and Game Boards often contain both prior substitution and hallucination. Flags is different again. It has the largest correct share and margins close to zero, but it also contains substantial hallucination. This fits the earlier interpretation of Flags as

Main Split: Low-Entropy Decisions Often Prefer the Prior

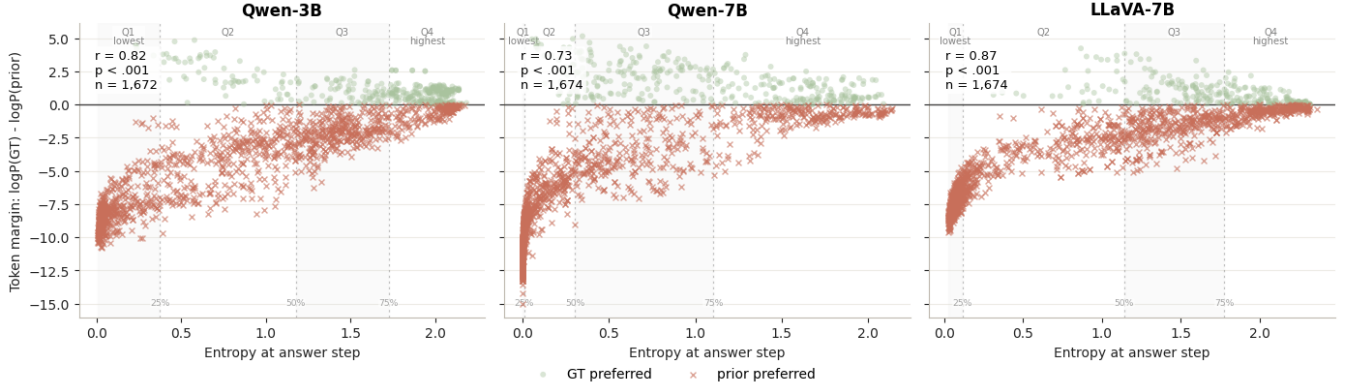


Figure 5: Main Split token-level entropy and margin by model. Each point is one token-scored row. The margin is defined as $\log P(\text{GT}) - \log P(\text{prior})$, so points below zero indicate prior preference. Across all three models, low-entropy decisions cluster in the negative-margin region, showing that the most confident answer-token decisions often favour the prior rather than the visible ground truth. Higher-entropy rows move closer to zero or into the GT-preferred region, suggesting that visual evidence competes with the prior mainly when the model is less certain. Dashed vertical lines mark entropy quartiles, and the reported r values show Pearson correlations with 95% confidence intervals.

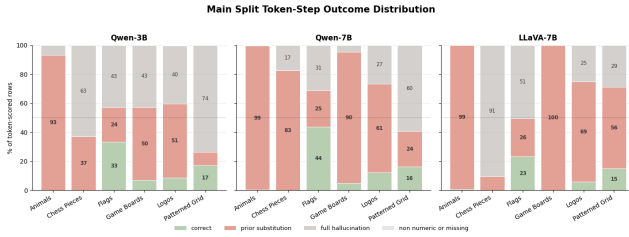


Figure 6: Outcome distribution per domain and model under token-step scoring. Each bar shows the proportion of samples where the model answered correctly (green), produced the bias-aligned answer (prior substitution, pink), or generated a response matching neither candidate (full hallucination, grey).

a weaker prior domain with a noisier visual counting task.

The two modes have different implications for intervention. Prior substitution suggests a breakdown in visual grounding before or during early token selection, where canonical associations dominate the output distribution almost immediately. Full hallucination instead reflects a failure under uncertainty, where the model does not confidently commit to the prior but still fails to stabilise around the visually correct answer. Methods that explicitly suppress prior-driven preferences, such as VCD, should therefore have greater traction on prior substitution than on full hallucination. This distinction motivates the intervention analysis in the following section.

VCD as a Diagnostic Probe

The failure modes identified in Section 5 suggest that not all forms of bias should respond equally to intervention. If

prior substitution reflects learned associations dominating the output distribution before visual evidence can meaningfully influence generation, suppressing language priors at inference time may have limited effect. Conversely, if full hallucination reflects uncertainty during the decision step itself, contrastive decoding may fail for reasons unrelated to prior dominance. Instead, the model may fail to converge on the visually grounded answer.

VCD is applied here not as a mitigation strategy, but as a diagnostic probe. Rather than asking only whether VCD improves accuracy, the analysis examines where it shifts probability mass, where it fails to do so, and whether those patterns align with the failure modes identified in RQ1.

Across all three models and both distortion types, VCD changes gt-preference rates only modestly. Under Gaussian noise, the proportion of samples preferring the gt answer shifts from 19.4% to 18.8% for Qwen2.5-VL-3B, 15.3% to 18.8% for Qwen2.5-VL-7B, and 16.9% to 19.1% for LLaVA-OV-7B. The proportion of samples actively helped by VCD, defined as cases where the margin shifts in the correct direction, reaches 41.4%, 43.0%, and 40.2% respectively. All three values remain below the 50% threshold that would indicate consistent net benefit across the dataset. Blur produces uniformly weaker results, with helped rates of 35.7%, 27.4%, and 24.6%.

The gap between noise and blur is consistent across all three models and is itself informative. Gaussian noise at $\sigma = 100$ corresponds to the distortion regime VCD was originally designed around. The underlying assumption is that degrading the image with structured noise suppresses visual detail and increases reliance on language priors, making the distorted run a useful approximation of prior-driven generation. Blur does not degrade the image in the same way, and the substantially weaker results under blur sug-

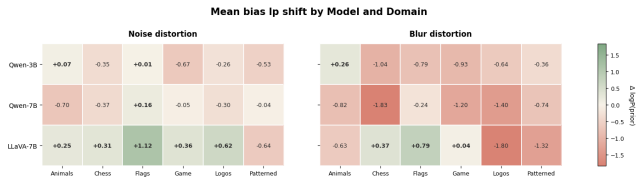


Figure 7: Mean `bias_lp_shift` per model and domain under noise (left) and blur (right). Green cells indicate the distortion increases bias log-probability as VCD assumes; red cells indicate the assumption fails. Flags is the only domain with consistently positive shifts across both distortions.

gest that the distortion choice is not interchangeable. A contrastive method calibrated to one degradation regime does not transfer cleanly to another, raising broader questions about the robustness of VCD’s core assumption in counterfactual visual reasoning settings.

VCD’s Assumption Check VCD’s corrective logic rests on a specific assumption. Replacing the original image with a distorted version should increase the model’s reliance on language priors, making the contrast between original and distorted predictions informative for isolating visually grounded information (Leng et al. 2024). In our setting, this implies that the bias candidate should receive higher probability under the distorted image than under the original image. The quantity `bias_lp_shift`, defined as the change in bias log-probability between the distorted and original images, directly tests this assumption. A positive shift indicates that the distortion increases prior preference as intended, while a negative shift indicates that the distortion fails to activate the prior in the expected direction.

Across most conditions, the assumption does not hold (Figure 7). Averaged across all six model and distortion combinations, `bias_lp_shift` is negative in five out of six settings, and the proportion of samples with a positive shift ranges from 17.9% to 58.7%. Under blur, the assumption holds in fewer than 37% of samples for all three models, with Qwen2.5-VL-7B performing worst at 17.9%. Even under Gaussian noise, the distortion regime VCD was originally designed around, the assumption holds in only 31.3% and 38.8% of samples for Qwen2.5-VL-7B and Qwen2.5-VL-3B respectively. LLaVA-OV-7B under noise is the only condition where the assumption holds in a majority of samples (58.7%), yet even there VCD improves the margin in only 40.2% of cases. The assumption therefore appears necessary but far from sufficient.

This pattern connects directly to the failure modes identified in RQ1. In domains dominated by prior substitution, particularly Animals and Chess Pieces, the bias log-probability is already near its ceiling under the original image. Distorting the input leaves little room for the bias probability to increase further, making negative `bias_lp_shift` values common by default. The prior is not being amplified by the distortion because it is already strongly preferred under the original image. As a result, manipulating the visual signal has limited effect on the underlying probability pref-

erence, reducing the effectiveness of VCD.

Visually-Gated versus Upstream Bias Against this backdrop of largely ineffective intervention, Flags emerges as a consistent exception. As shown in Table 1, VCD under Gaussian noise helps 62–65% of Flags samples across all three models, making it the only domain where helped rates consistently exceed the 50% threshold. The corresponding `bias_lp_shift` values are positive on average, confirming that the distortion increases reliance on the canonical prior in the manner VCD assumes. The method succeeds here precisely because the bias remains contingent on the visual signal, and suppressing that signal shifts the distribution toward the memorized prior.

The reason Flags behaves differently from Animals and Chess Pieces returns to the prior-strength interpretation developed in RQ1. Flag configurations, such as the number of stars, stripes, or colours, do not collapse into a single overwhelmingly dominant association during pretraining in the same way as four-legged animals or standard chess piece counts. The preference for the biased answer in Flags therefore appears only partially upstream and remains dependent on properties of the image itself. When VCD suppresses the visual signal through distortion, the prior is genuinely activated, `bias_lp_shift` becomes positive, and the contrastive adjustment has meaningful probability mass to operate on.

These results motivate a distinction between two forms of bias. *Upstream bias*, exemplified by Animals and Chess Pieces, is driven by canonical associations strong enough to dominate the output distribution largely independently of the visual input. VCD fails in these settings because the prior is not being newly activated by the distortion; it is already dominant under the original image. *Visually-gated bias*, exemplified by Flags, depends more directly on the interaction between the image and the learned prior. In these cases, the distorted input produces the contrastive shift that VCD assumes, allowing the method to meaningfully redirect probability mass toward the visually grounded answer. The distinction between these two bias types therefore appears to arise not from architecture or decoding strategy alone, but from how strongly a domain’s learned associations have been consolidated during pretraining.

Table 1: Percentage of samples helped by VCD ($\alpha = 1.0$) per domain, model, and distortion type. Values above 50% indicate net benefit. Flags is the only domain consistently exceeding this threshold across all models and distortions.

	Qwen-3B		Qwen-7B		LLaVA-7B	
Domain	Noise	Blur	Noise	Blur	Noise	Blur
Animals	49.3	41.2	32.1	29.3	26.6	6.2
Chess Pieces	40.6	29.5	34.4	5.6	26.0	4.9
Flags	62.1	64.2	65.0	56.3	62.5	61.3
Game Boards	44.0	39.9	55.4	38.7	48.2	33.3
Logos	40.6	25.4	54.6	26.6	52.7	20.8
Patterned Grid	14.0	22.6	31.8	17.9	39.3	45.5
Overall	41.4	35.7	43.0	27.4	40.2	24.6

Alpha Sensitivity The VCD objective is controlled by a scaling parameter α , which determines the strength of the contrastive adjustment applied to the output distribution. If VCD is successfully suppressing a visually-gated prior, increasing α should strengthen the shift toward the visually grounded answer. Conversely, if the distorted run does not meaningfully activate the prior, changing α should have little effect because there is no useful contrastive signal to amplify.

The results strongly support the latter interpretation. As shown in Figure 8, varying α from 0.5 to 2.0 produces only minor changes in gt-preference rates across all three models and both distortions. Under Gaussian noise, Qwen2.5-VL-7B shifts from 18.1% at $\alpha = 0.5$ to 19.8% at $\alpha = 2.0$, while LLaVA-OV-7B under blur changes from 18.7% to 19.3%. Across all conditions, the total variation remains within approximately four percentage points.

The flat trajectories indicate that VCD’s behaviour is largely insensitive to the precise strength of the contrastive adjustment within this range. Even in Flags, where VCD exhibits meaningful traction, larger α values do not substantially improve performance. This suggests that the dominant factor is not the magnitude of the contrastive update itself, but whether the distorted input successfully activates a usable prior signal in the first place. Once that condition is met, increasing α yields only limited additional benefit.

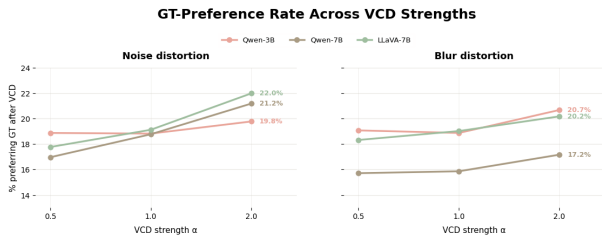


Figure 8: GT-preference rate across $\alpha \in \{0.5, 1.0, 2.0\}$ for all three models under noise (left) and blur (right). The largely flat trajectories indicate that VCD’s effectiveness is relatively insensitive to the strength of the contrastive adjustment within this range.

6 Discussion

This study shows that counterfactual VLM bias is not only an accuracy drop on harder images. The stronger finding is that, when visual evidence conflicts with learned expectations, the prior-consistent answer is often already favoured inside the model distribution. This extends VLMBias from a benchmark-level observation to a probability-level account of the failure (Vo et al. 2025). The models do not simply generate the wrong answer at the surface level. In many cases, the wrong answer has already become the preferred answer before the final response is evaluated.

What Probability-Level Analysis Adds

Earlier VQA work showed that models can exploit language regularities in the question rather than relying on the image

(Goyal et al. 2017). Here, the shortcut is not mainly introduced by wording. It is activated by the familiar visual subject itself. Figure 3 and Figure 4 show this most clearly for Animals and Chess Pieces. In these domains, the model recognizes a familiar object or structure, retrieves the expected count, and often fails to revise that expectation when the image contradicts it.

The domain pattern suggests that there is no single bottleneck. Animals are the cleanest case of prior override. The Original Split shows that the models can count animal legs when the image agrees with the prior, while the Main Split shows that this ability collapses when the count becomes unusual. This supports a use-based explanation. The visual signal may be present, but it does not control the answer. This interpretation is consistent with the VLMBias linear-probe result, where vision encoder features contain useful leg-count information even though the full VLM defaults to the biased answer (Vo et al. 2025). It also fits the broader view that VLMs may encode relevant visual evidence but fail to use it during answer selection (Liu et al. 2025).

We observe a less clean pattern for the other domains. Flags, Logos, Game Boards, and Patterned Grid require counting repeated or local details, which previous work identifies as difficult for vision-language systems (Paiss et al. 2023; Campbell et al. 2025; Tong et al. 2024). This helps explain why Flags do not behave like Animals. Flags are familiar, but the task often requires exact counting of stars or stripes. The weaker prior signal in Flags should therefore not be interpreted as reliable visual grounding. It more likely reflects a mixture of weaker prior dominance and ordinary counting difficulty.

PMI correction helps separate these explanations. In Animals, PMI changes little, which strengthens the interpretation of semantic prior override. In Logos and Patterned Grid, especially for Qwen2.5-VL-3B, PMI improves ground-truth preference substantially. This means that the raw margins in these domains are partly shaped by context-free number preference or candidate-set effects. For Logos, this is plausible because many questions involve common small counts linked to stripes, circles, curves, or points. For Patterned Grid, the interpretation is different. The images are constructed from scratch and do not correspond to familiar Internet concepts (Vo et al. 2025). A large PMI shift there suggests that the raw margin is not measuring a clean memorized semantic prior. It also reflects generic numeric bias and instability in the candidate comparison.

Patterned Grid is the most useful boundary case in the benchmark. Unlike Animals or Chess Pieces, it does not test whether the model remembers a familiar world fact. It tests whether the model counts a local anomaly against a strong surrounding pattern. The benchmark itself defines the expected bias as the pattern-consistent value inferred from neighbouring cells, while the ground truth is the actual modified count in the target cell. In our sequence data, these two candidates are always only one count apart. This means that a margin close to zero does not necessarily imply strong visual grounding. It may simply mean that neither candidate is a stable attractor. The outcome analysis supports this interpretation, since Patterned Grid is dominated by full halluci-

nation for some models rather than clean prior substitution. The model often does not settle on the visual answer or the pattern-consistent answer. It produces a third number.

When Hallucination Appears

Prior hallucination work often describes image-inconsistent object generation, where models produce plausible objects that are frequent or likely to co-occur with the scene (Li et al. 2023). Our results show a numeric analogue. When grounding is weak, the model may generate a plausible number rather than a plausible object. This is related to the VCD view that visual uncertainty can amplify reliance on language priors and statistical bias (Leng et al. 2024). However, our results also show that hallucination is not the same as prior substitution. Prior substitution means the model selects the expected canonical answer. Full hallucination means the model leaves both the ground truth and the predefined prior. These failures require different explanations and likely different interventions.

This is also why the positive Qwen2.5-VL-3B margin on Game Boards should be treated carefully. It is the only clearly positive Main Split margin that is not near zero, but a positive margin only means that the ground-truth candidate is scored above the predefined prior candidate. It does not guarantee that the generated answer is correct. Figure 6 shows why this matters. Models can leave both candidates and generate another wrong number. Margins are useful for measuring competition between two selected answers, but they cannot fully capture open-ended numeric instability.

The entropy results help explain why these errors are difficult to correct. Figure 5 shows that low entropy on the Main Split often favours the prior. This reverses the Original Split control (See Appendix C.3), where low entropy usually favours the ground truth. Confidence becomes unreliable under counterfactual conflict. It can indicate that the model has already committed to the familiar answer. This helps explain why simple prompting or double-checking may have limited effect. If the first answer token has already collapsed toward the prior, later reasoning may only justify the wrong commitment.

VCD confirms that not all bias is located in the same place. The method assumes that a distorted image increases reliance on language priors, creating a distribution that can be subtracted (Leng et al. 2024). However, our result does not fully align with that. For Animals and Chess Pieces, the prior appears already active under the original image. Distortion cannot reveal a hidden prior because the prior is already dominating the real-image distribution. Flags are the main exception. Table 1 shows that VCD helps Flags more consistently, suggesting that flag bias is more visually gated. The flat α curves in Figure 8 support the same interpretation. The issue is not mainly the strength of the contrastive update. It is whether the distorted image creates a valid correction signal.

The comparison between Qwen2.5-VL-3B and Qwen2.5-VL-7B also suggests that scale alone is not sufficient. The larger Qwen model often shows stronger negative margins, which resembles inverse-scaling cases where larger models perform worse when a learned shortcut conflicts with the in-

tended task (McKenzie et al. 2024). This should be treated as suggestive because only two Qwen sizes are compared. Still, it points to an important possibility. Larger models may retrieve canonical knowledge more reliably without becoming better at revising that knowledge when the image contradicts it.

Limitations

This study is limited by the number and type of models evaluated. Although the three models differ in architecture and scale, they all use the Qwen2 tokenizer. This shared tokenizer helps make token-level comparisons more consistent, but it also means that tokenization is not an independent source of variation in the analysis. A broader model set would be needed to determine whether the observed failure modes and apparent scale effects generalise across alternative tokenizers, vision encoders, and training regimes.

The probability framework also has limits. Candidate scoring is useful because it gives a controlled comparison between the ground truth and the predefined prior answer. It is less suited to errors where the model generates a third answer outside both candidates. This is especially relevant for domains such as Patterned Grid and Game Boards, where full hallucination and numeric instability play a larger role. The token-step analysis is similarly restricted to single-token numeric answers. This makes the decision point easy to interpret, but it excludes multi-token responses and does not capture all forms of open-ended generation.

The intervention analysis should also be interpreted cautiously. VCD was tested with Gaussian noise and blur, together with a limited range of contrastive strengths. The results show that VCD is sensitive to the distortion used, which means that other distortions could produce different conclusions. More targeted interventions, such as object masking, local blur around the edited region, cropping, or background removal, would be needed before the distinction between visually gated and upstream bias can be treated as a general taxonomy.

There are also dataset-level limitations. The analysis relies on the VLMBias benchmark, which is carefully constructed but still contains controlled counterfactual images rather than natural user queries. Some counterfactual edits may introduce artifacts, and overlap between familiar benchmark subjects and model pretraining data cannot be ruled out. In addition, the study covers six domains and focuses mainly on counting and recognition. The findings describe prior-driven behaviour under controlled counterfactual conditions, but they should not be overgeneralised to all multi-modal reasoning tasks.

Finally, some domains are cleaner diagnostic cases than others. Animals provide a strong control because original animal counting is solved reliably, while Patterned Grid has no direct original counterpart because the images are created from scratch. This makes Patterned Grid useful for studying hallucination and local counting failure, but less suitable for claims about memorised semantic priors. The interpretation of each domain should remain tied to the strength of its control condition.

7 Conclusion

This paper investigated prior-driven behavior in vision-language models during counterfactual visual reasoning. Using probability-level diagnostics across three open-source VLMs, we found that biased answers are frequently preferred over visually correct alternatives before the final response is produced. Sequence-level analysis showed consistent probability preference for expected bias candidates across domains, while token-level analysis revealed that these preferences are often associated with low-entropy, high-confidence decisions.

The analysis also revealed recurring patterns of prior substitution and full hallucination. In prior substitution, models favor a canonical answer despite contradictory visual evidence. In full hallucination, generation diverges toward a third unsupported response. Together, these findings suggest that counterfactual failures are not limited to simple answer selection errors, but reflect broader failures of visual grounding.

We further investigate these probability preferences by evaluating Visual Contrastive Decoding as both an inference-time intervention and a diagnostic probe. The VCD results revealed substantial variation across domains. The strongest and most consistent improvements were observed for Flags, whereas limited gains were observed elsewhere. This variation suggests that not all prior-driven preferences are equally dependent on visual evidence. Some remain responsive to changes in the visual signal, while others persist despite visual intervention.

Overall, our findings show that analyzing probability distributions provides insight that is not visible from final-answer accuracy alone. By examining model preferences before answer generation, we find that counterfactual failures are often characterized by strong prior-consistent probability preferences, and that the effectiveness of VCD depends on whether those preferences remain sensitive to visual evidence. Future work may extend this analysis to additional model families, counterfactual benchmarks, and alternative interventions for improving visual grounding.

8 Acknowledgments

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References

Bai, J.; Bai, S.; Yang, S.; Wang, S.; Tan, S.; Wang, P.; Lin, J.; Zhou, C.; and Zhou, J. 2023. Qwen-VL: A Versatile Vision-Language Model for Understanding, Localization, Text Reading, and Beyond. *arXiv preprint arXiv:2308.12966*.
Campbell, D.; Rane, S.; Giallanza, T.; Sabbata, N. D.; Ghods, K.; Joshi, A.; Ku, A.; Frankland, S. M.; Griffiths, T. L.; Cohen, J. D.; and Webb, T. W. 2025. Understanding the

Limits of Vision Language Models Through the Lens of the Binding Problem. *arXiv:2411.00238*.

Deng, A.; Cao, T.; Chen, Z.; and Hooi, B. 2025. Words or Vision: Do Vision-Language Models Have Blind Faith in Text? *arXiv:2503.02199*.

Fu, C.; Chen, P.; Shen, Y.; Qin, Y.; Zhang, M.; Lin, X.; Yang, J.; Zheng, X.; Li, K.; Sun, X.; Wu, Y.; Ji, R.; Shan, C.; and He, R. 2025. MME: A Comprehensive Evaluation Benchmark for Multimodal Large Language Models. *arXiv:2306.13394*.

Goyal, Y.; Khot, T.; Summers-Stay, D.; Batra, D.; and Parikh, D. 2017. Making the V in VQA Matter: Elevating the Role of Image Understanding in Visual Question Answering. *arXiv:1612.00837*.

Guan, T.; Liu, F.; Wu, X.; Xian, R.; Li, Z.; Liu, X.; Wang, X.; Chen, L.; Huang, F.; Yacooob, Y.; Manocha, D.; and Zhou, T. 2024. HallusionBench: An Advanced Diagnostic Suite for Entangled Language Hallucination and Visual Illusion in Large Vision-Language Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 14375–14385.

Guo, C.; Pleiss, G.; Sun, Y.; and Weinberger, K. Q. 2017. On Calibration of Modern Neural Networks. *arXiv:1706.04599*.

Huang, Q.; Dong, X.; Zhang, P.; Wang, B.; He, C.; Wang, J.; Lin, D.; Zhang, W.; and Yu, N. 2024. OPERA: Alleviating Hallucination in Multi-Modal Large Language Models via Over-Trust Penalty and Retrospection-Allocation. *arXiv:2311.17911*.

Huo, F.; Xu, W.; Zhang, Z.; Wang, H.; Chen, Z.; and Zhao, P. 2025. Self-Introspective Decoding: Alleviating Hallucinations for Large Vision-Language Models. *arXiv:2408.02032*.

Kadavath, S.; Conerly, T.; Askell, A.; Henighan, T.; Drain, D.; Perez, E.; Schiefer, N.; Hatfield-Dodds, Z.; DasSarma, N.; Tran-Johnson, E.; Johnston, S.; El-Showk, S.; Jones, A.; Elhage, N.; Hume, T.; Chen, A.; Bai, Y.; Bowman, S.; Fort, S.; Ganguli, D.; Hernandez, D.; Jacobson, J.; Kernion, J.; Kravec, S.; Lovitt, L.; Ndousse, K.; Olsson, C.; Ringer, S.; Amodei, D.; Brown, T.; Clark, J.; Joseph, N.; Mann, B.; McCandlish, S.; Olah, C.; and Kaplan, J. 2022. Language Models (Mostly) Know What They Know. *arXiv:2207.05221*.

Kolari, A.; Khojasteh, M.; Jiang, Y.; den Hengst, F.; and Ilievski, F. 2026. A Study of Commonsense Reasoning over Visual Object Properties. *arXiv:2508.10956*.

Kuhn, L.; Gal, Y.; and Farquhar, S. 2023. Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation in Natural Language Generation. *arXiv:2302.09664*.

Lee, K.-i.; Kim, M.; Yoon, S.; Kim, M.; Lee, D.; Koh, H.; and Jung, K. 2025. VLind-Bench: Measuring Language Priors in Large Vision-Language Models. In *Findings of the Association for Computational Linguistics: NAACL 2025*, 4129–4144. Association for Computational Linguistics.

Leng, S.; Zhang, H.; Chen, G.; Li, X.; Lu, S.; Miao, C.; and Bing, L. 2024. Mitigating Object Hallucinations in Large Vision-Language Models through Visual Contrastive Decoding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 13872–13882.

- Li, B.; Zhang, Y.; Guo, D.; Zhang, R.; Li, F.; Zhang, H.; Zhang, K.; Li, Y.; Liu, Z.; and Li, C. 2024. LLaVA-OneVision: Easy Visual Task Transfer. *arXiv preprint arXiv:2408.03326*.
- Li, Y.; Du, Y.; Zhou, K.; Wang, J.; Zhao, W. X.; and Wen, J.-R. 2023. Evaluating Object Hallucination in Large Vision-Language Models. *arXiv:2305.10355*.
- Liu, H.; Li, C.; Wu, Q.; and Lee, Y. J. 2023. Visual Instruction Tuning. In *Advances in Neural Information Processing Systems*, volume 36, 34892–34916.
- Liu, Z.; Chen, Z.; Liu, H.; Luo, C.; Tang, X.; Wang, S.; Zeng, J.; Dai, Z.; Shi, Z.; Wei, T.; Dumoulin, B.; and Tong, H. 2025. Seeing but Not Believing: Probing the Disconnect Between Visual Attention and Answer Correctness in VLMs. *arXiv:2510.17771*.
- Luo, T.; Cao, A.; Lee, G.; Johnson, J.; and Lee, H. 2025. Probing Visual Language Priors in VLMs. *arXiv preprint arXiv:2501.00569*.
- McKenzie, I. R.; Lyzhov, A.; Pieler, M.; Parrish, A.; Mueller, A.; Prabhu, A.; McLean, E.; Kirtland, A.; Ross, A.; Liu, A.; Gritsevskiy, A.; Wurgajt, D.; Kauffman, D.; Recchia, G.; Liu, J.; Cavanagh, J.; Weiss, M.; Huang, S.; Droid, T. F.; Tseng, T.; Korbak, T.; Shen, X.; Zhang, Y.; Zhou, Z.; Kim, N.; Bowman, S. R.; and Perez, E. 2024. Inverse Scaling: When Bigger Isn’t Better. *arXiv:2306.09479*.
- Paiss, R.; Ephrat, A.; Tov, O.; Zada, S.; Mosseri, I.; Irani, M.; and Dekel, T. 2023. Teaching CLIP to Count to Ten. *arXiv:2302.12066*.
- Rahmanzadehgervi, P.; Bolton, L.; Taesiri, M. R.; and Nguyen, A. T. 2025. Vision language models are blind: Failing to translate detailed visual features into words. *arXiv:2407.06581*.
- Sharma, P.; Shaham, T. R.; Baradad, M.; Fu, S.; Rodriguez-Munoz, A.; Duggal, S.; Isola, P.; and Torralba, A. 2024. A Vision Check-up for Language Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 14410–14419.
- Sutskever, I.; Vinyals, O.; and Le, Q. V. 2014. Sequence to Sequence Learning with Neural Networks. In *Advances in Neural Information Processing Systems*, volume 27.
- Tong, S.; Liu, Z.; Zhai, Y.; Ma, Y.; LeCun, Y.; and Xie, S. 2024. Eyes Wide Shut? Exploring the Visual Shortcomings of Multimodal LLMs. *arXiv:2401.06209*.
- Vo, A.; Nguyen, K.; and collaborators. 2025. VLMs Are Biased. Associated with the paper “Vision Language Models are Biased”.
- Vo, A.; Nguyen, K.-N.; Taesiri, M. R.; Dang, V. T.; Nguyen, A. T.; and Kim, D. 2025. Vision Language Models are Biased. *arXiv preprint arXiv:2505.23941*.
- Wang, S.; et al. 2025. Qwen2.5-VL Technical Report. *arXiv preprint arXiv:2502.13923*.
- Wang, X.; Pan, J.; Ding, L.; and Biemann, C. 2024. Mitigating Hallucinations in Large Vision-Language Models with Instruction Contrastive Decoding. *arXiv:2403.18715*.
- Zhao, Z.; Wallace, E.; Feng, S.; Klein, D.; and Singh, S. 2021. Calibrate Before Use: Improving Few-Shot Performance of Language Models. In *International Conference on Machine Learning*.

A Phase 1 Pipeline Overview

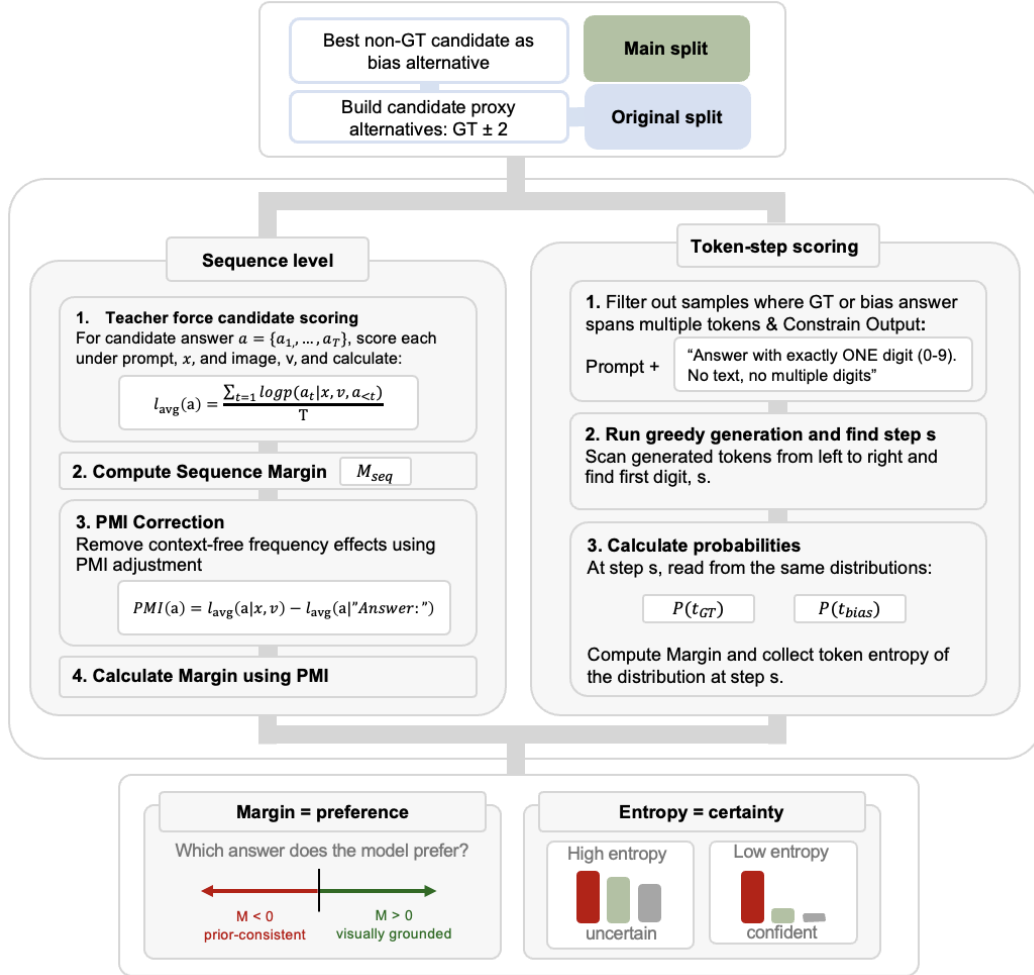


Figure 1: Overview of the Phase 1 probability-level analysis pipeline. The sequence-level branch scores complete candidate answers under teacher forcing and compares them using log-probability margins, with PMI correction used to reduce answer-prior effects. The token-level branch constrains generation to a single-digit answer and directly inspects the model’s first answer-token distribution. In both branches, margin direction measures candidate preference, while entropy measures prediction certainty.

B Phase 2 Pipeline Overview

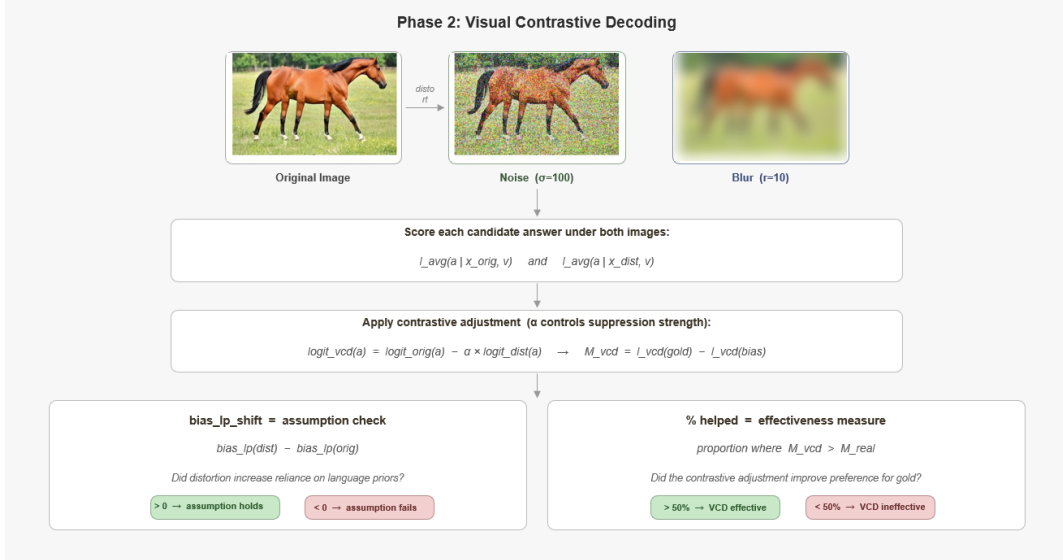


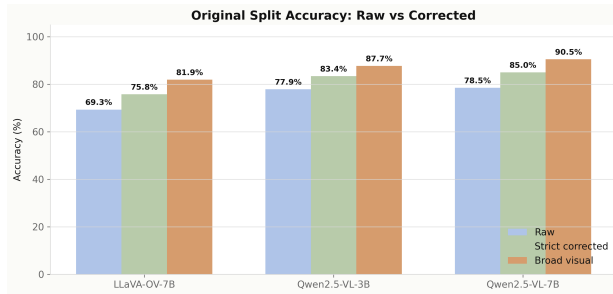
Figure 2: Overview of the Visual Contrastive Decoding (VCD) diagnostic pipeline. For each image, a counterfactual version is produced using either Gaussian noise ($\sigma = 100$) or blur ($r = 10$). Candidate answers are then scored under both the original and distorted visual inputs. The contrastive adjustment subtracts distorted-image logits, scaled by α , from the original-image logits in order to suppress prior-driven probability mass. Two derived diagnostics are used in the analysis: `bias_lp_shift`, which tests whether distortion activates language priors as assumed by VCD, and `% helped`, which measures whether the adjusted margin improves over the baseline margin.

C Original Split Control and Sanity Checks

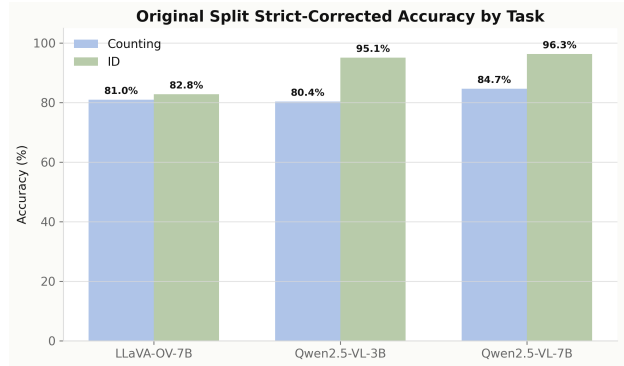
The Original Split is used as a control condition for visual competence. Unlike the Main Split, it contains unedited images in which the visible evidence and the canonical prior are aligned.

C.1 Overall Original Split Accuracy

Figure 3 reports Original Split accuracy by model and by task. Raw accuracy is based on exact automatic matching, while corrected accuracy incorporates manual review of visually valid aliases. The correction has the largest effect on identification questions, where semantically correct answers may differ lexically from the dataset label. Counting questions are largely unaffected because numeric answers are evaluated by exact equality.



(a) Accuracy by model under raw, strict-corrected, and broad-corrected evaluation.



(b) Strict-corrected accuracy by task type.

Figure 3: Original Split accuracy under raw and corrected evaluation. Broad visual correction primarily affects identification questions, where visually correct responses can be phrased differently from the dataset label. Counting accuracy is comparatively stable because numeric answers are evaluated by exact equality.

C.2 Task- and Domain-Level Accuracy

Figure 4 separates Original Split performance by task type and domain. These subgroup results should be interpreted as diagnostic evidence rather than stable estimates, because the Original Split is small and unevenly distributed across domains. Animals and Logos contain more examples than Chess Pieces and Game Boards, making percentages in smaller domains more sensitive to individual samples.

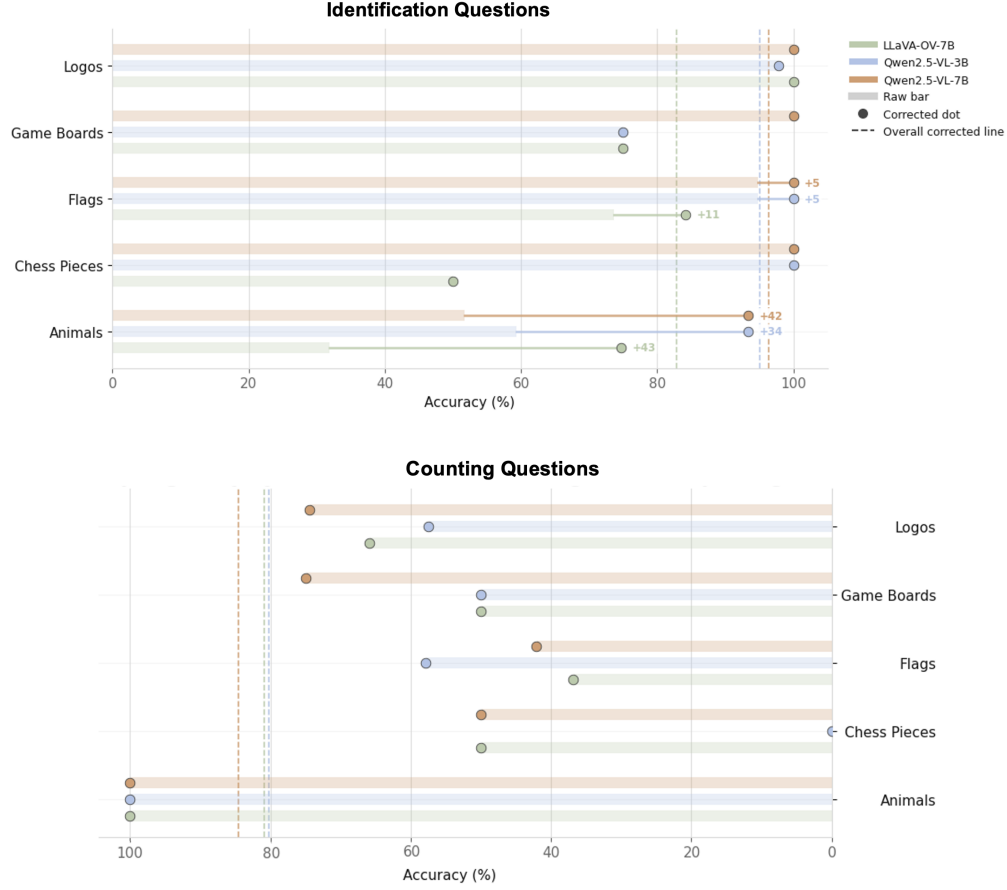


Figure 4: Original Split subgroup accuracy for identification and counting questions. Pale bars show raw automatic matching, filled dots show corrected accuracy, and dashed vertical lines show each model’s overall corrected accuracy for the corresponding question type. Correction has little effect on counting because numeric answers are evaluated by exact equality, while identification benefits from conservative alias correction and broader visual-category matching. Because the split is small and uneven across domains, these subgroup results are used as qualitative diagnostics rather than as stable performance estimates.

C.3 Original Split Probability-Control Diagnostics

The probability-control diagnostics verify that the probability-level methods behave as expected when no counterfactual conflict is present. On Original Split numeric rows, all three models typically prefer the ground-truth token over the proxy alternative. Token-level and sequence-level margins are also strongly aligned on matched rows, supporting the use of sequence-level candidate scoring as a broader diagnostic.

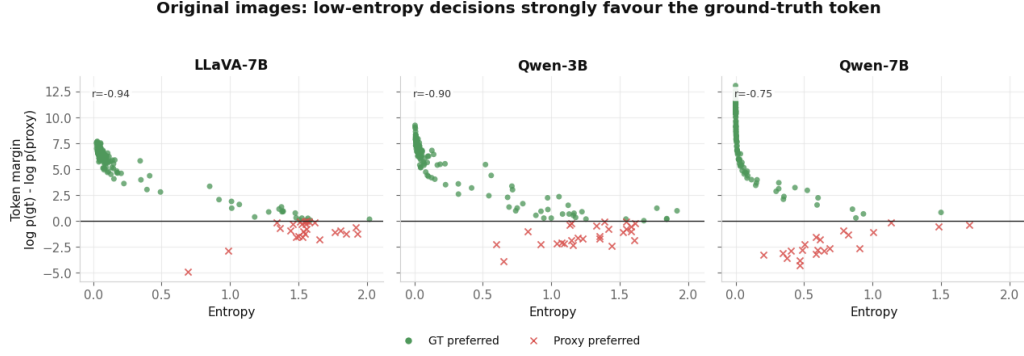


Figure 5: Entropy-margin control on the Original Split. Low-entropy decisions tend to favour the ground-truth token over the proxy alternative, indicating that model confidence is visually grounded when the image is unedited.

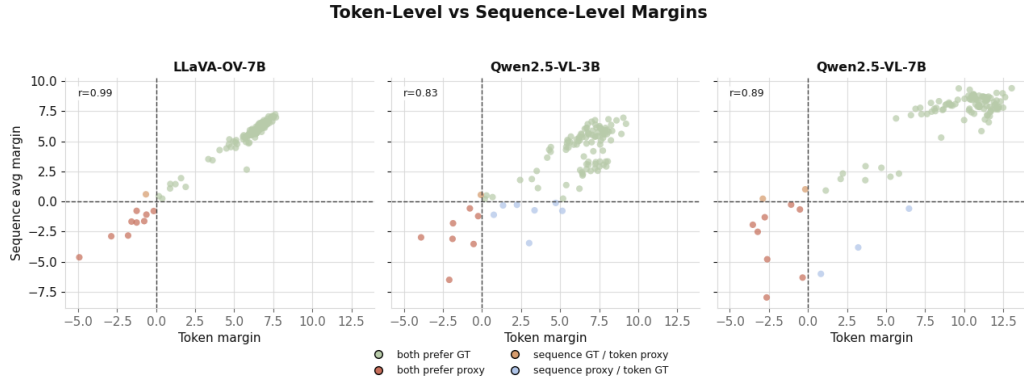


Figure 6: Agreement between token-level and sequence-level margins on Original Split numeric rows. Most points fall in the ground-truth preference quadrant, showing that the two probability diagnostics are aligned in the control setting.

D Token-vs-Sequence Agreement on the Main Split

Figure 7 compares token-level and sequence-level margins on the Main Split. This diagnostic checks whether the single-token probability probe and the full-sequence candidate-scoring probe identify similar preference directions under counterfactual conflict. Agreement between the two measures supports the interpretation that sequence-level scoring captures the same underlying model preference measured by the more constrained token-level analysis.

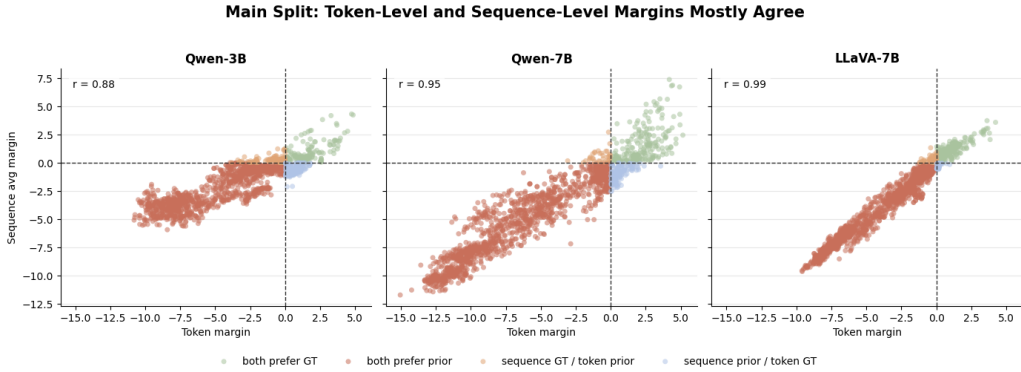


Figure 7: Agreement between token-level and sequence-level margins on the Main Split. Each point compares the model’s preference direction under the constrained first-token diagnostic with its preference direction under full-sequence candidate scoring. Points in matching quadrants indicate agreement between the two probability-level probes, while off-diagonal points indicate cases where token-level and sequence-level preferences diverge.